Hamza Basem Mohamed Ahmed Eldafrawy

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PROFILE

Computer and Systems Engineering student (Class of 2027, GPA: 3.62) with experience in backend development, distributed systems, machine learning, and data-driven architectures. Built foundational projects involving socket programming, recommendation systems, and software architecture using Python, C#, Java, and C++. Dedicated to a first-principles engineering mindset, combining robust backend engineering with intelligent systems to build scalable, real-world applications.

EDUCATION

Ain Shams University — Faculty of Engineering Cairo, Egypt
B.Sc. in Computer and Systems Engineering Expected Graduation: 2027

- **Cumulative GPA:** 3.62/4.00
- **Relevant Coursework:** Deep Learning, Data Structures & Algorithms, Operating Systems, Parallel and Distributed Systems, Databases, Fundamentals of Big Data Analysis.

TECHNICAL SKILLS

- **Programming Languages:** Python, C#, SQL, C++, C, Java
- **Data & Machine Learning:** pandas, NumPy, scikit-learn, Basic NLP, Power BI, Data Visualization
- **Backend & Systems:** Linux/Unix OS, Socket Programming, ADO.NET, Windows Forms, Qt 6, Git
- **Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Relational Database Design, Distributed Architecture

PROJECTS PORTFOLIO

Online Marketplace Backend — C#, Socket Programming

- Architected an online marketplace platform for a Parallel & Distributed Systems course alongside a backend team.
- Implemented core network communication utilizing raw socket programming and applied SOLID principles to establish a deep understanding of distributed systems and client-server concurrency.

Bank Management System — C#, SQL Server, ADO.NET, Windows Forms

- Designed a relational database architecture in SQL Server (8 interconnected tables) and integrated it with a multi-form Windows Forms desktop application.
- Implemented a centralized data access layer using ADO.NET to manage CRUD operations, parameterized queries, and basic transaction processing.

Vezeeta Healthcare Recommendation Engine — Python, pandas, Basic NLP

- Engineered an end-to-end data ingestion pipeline to scrape, clean, and preprocess over 17,000 complex, unstructured medical profiles into structured formats.
- Utilized foundational Machine Learning and basic NLP similarity matching to accurately map patient symptoms to the appropriate medical professionals.

XML Social Network Analyzer — C++, Qt 6

- Collaborated within an 8-member team using modular development practices to build a multi-feature desktop application.
- Implemented a social network analyzer utilizing graph algorithms and developed a JSON converter module enabling XML-to-JSON transformation.

Water Access Analytics Dashboard — SQL, Power BI

- Extracted and manipulated raw regional project data using complex SQL queries (MySQL).
- Developed a comprehensive, interactive Power BI dashboard to visualize findings and translate data into actionable business insights.

Micro Shell — C, Linux

- Implemented a Unix-like shell supporting command execution, I/O redirection, and variable parsing.
- Practiced low-level process management utilizing process forking (`fork()`, `execvp()`), file descriptors, and signal handling.

Online Language Teaching System — Java, OOP

- Led a 3-member team to build an interactive GUI-based learning platform, applying strict Object-Oriented Programming concepts (encapsulation, inheritance) to cleanly separate teacher, student, and lesson modules.

TRAINING & CONTINUOUS LEARNING

Machine Learning Specialization — Coursera (DeepLearning.AI) *06/2025 – 02/2026*

- Mastered core ML algorithms and foundational deep learning architectures (CNNs, GANs), building end-to-end ML pipelines emphasizing rigorous model evaluation.

Data Science Track — ALX - Africa (with ExploreAI Academy) *03/2025 – 04/2026*

- Applied clustering techniques and built classical ML pipelines (Decision Trees, SVM) using `scikit-learn`, alongside complex relational database design.

Full Stack Web Development (PHP) — National Telecommunication Institute (NTI)
- Completed intensive training on web development concepts, database connectivity, and backend integration.
- Led a student team to implement foundational backend logic including user authentication and basic CRUD operations.

Google IT Support — Coursera
- Completed foundational modules focusing on technical support, system administration, and computer networking.

Algorithmic Problem Solving (LeetCode) *Ongoing*
- Consistently solving daily algorithmic challenges to sharpen analytical thinking, algorithm optimization, and data structure proficiency.

Forward Program — McKinsey & Company
- Completed a competitive learning program focused on developing practical skills in business strategy, leadership, and structured problem-solving.

LEADERSHIP & ACTIVITIES

Section Coordinator — Faculty of Engineering, Ain Shams University
- Coordinated communication and academic support within the section across multiple semesters.
- Assisted colleagues with coursework-related challenges, reports, and scheduling conflicts.
- Helped organize commitments and facilitated collaboration between students when needed.

Student Union Member — Ain Shams University
- Participated in planning and organizing student activities and events.
- Assisted in coordinating logistics and student engagement initiatives.

LANGUAGES

- **Arabic:** Native **English:** Professional Proficiency